# Pilot Research Summary on the Impact of Sonorium on the Individual

Collaborators: The Svaram Team, especially Lukas Weick, Priyanjali Das, Astrid Slegten, Aurelio C. Hammer, Timon Berger, Joseph Schmidlin, Helena Becker

**Study Objective**

The study aimed to evaluate the null hypothesis regarding the impact of "Sonorium" on the physiological and psychological parameters of an individual.

The other main objective was to gather experience on how to conduct research in the Sonorium and see if there is any promising direction for future investigations.

**Introduction/Background**



Sonorium sessions are a form of vibroacoustic therapy that uses a sound bed and other instruments like a 64Hz gong and a collection of untuned and tuned instruments to create an integral sound environment in a more or less pre-established order. The research enriches concepts from integral psychology, the theories of Pancha Kosha and Panchamahabhuta from Yogic science, and Chinese medicine.

**Methodology:**

**Participants**

12 subjects, aged between 18 and 50, with no history of mental health conditions or psychoactive drug use, were selected. The subjects were recruited by posting a call for participants on the Whatsapp groups of Auroville. We gathered 24 interested candidates and from the applications we selected 6 female subjects and 6 male subjects. We also tried to have subjects from different nationalities. 6 of the subjects were from Indian and 6 were from non-Indian nationality.

**Design**

Tools such as self-assessment questionnaires, an electroencephalogram, pulse oximetry, a thermometer, and an Apple watch were employed.

The protocol included a standardized 30-minute Sonorium session, parameter measurements during the session, and pre- and post-session evaluations using the Bio-well and self-assessment questionnaires on perceived stress and sleep quality.

Psychological parameters were evaluated before the session and 1 month after the session.

We gathered qualitative data by recording the intake conversation before the session and recording an interview with the participants two times after the session.

**Parameters and Indicators Evaluated:**

Biometric parameters were evaluated, including:

- blood pressure
- heart rate variability (Apple watch SI)
- heart rate ( Pulsoxymeter)
- respiration rate ( Pulsoxymeter)
- brain electrical activity ( Bio Medical System)
- body temperature
- body movements
- eye movement  and facial expressions
- Bio-well machine was used to measure chakra-related parameters (Bio-well)

Psychological parameters measured included:

- Perceived stress scale
- Pittsburgh sleep quality index

Qualitative data that we collected:

- intake interview before the sound session
- report of the sound facilitator after the sound session
- follow-up interview after the session (3 days)
- follow-up interview after the session (1 month)

**Findings:**

**Pre Post Results:**

Evaluated parameters pre/post Sonorium were blood pressure, heart rate and respiration rate and body temperature.

Here, only the heart rate showed a significant decrease of on average 5 beats per min (p-value is 0.031, at a common significance level of 0.05).  All other parameters did not show significant changes.

For heart rate variability the paired t-test analysis reveals a significant difference between the pre and post values (p-value of 0.00).

Moreover, the average difference between pre and post measurements decreased by 21.48 ms from the pre to the post condition.

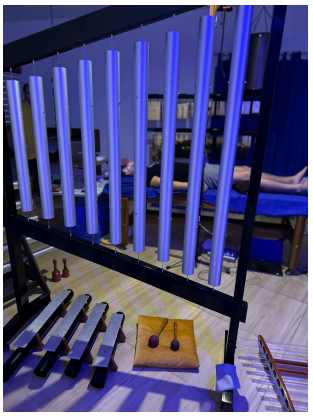
Bio-well machine findings analysis is still pending.

**EEG:**

At the end 2 data sets for EEG analysis were complete and were analyzed showing only 5.79% of beta waves, while having 73.2% of delta Waves, 11 % alpha waves and 9.83% theta waves.

| Total | Subject 1 | Subject 2 | Total |
|---|---|---|---|
| Alpha | 8.61% | 13.42% | 11.0% |
| Beta | 5.18% | 6.40% | 5.79 % |
| Delta | 79.58% | 66.83% | 73.17 % |
| Theta | 6.58% | 13.07% | 9.83% |



**Results psychological analysis:**

The following scores of the the subjects were collated
Perceived stress scale:

| **Participant** | **Pre** | **Post** |
|---|---|---|
| Participant 1 | 26 | 27 |
| Participant 2 | 23 | 22 |
| Participant 3 | 27 | 26 |
| Participant 4 | 29 | 31 |
| Participant 5 | 23 | 20 |
| Participant 6 | 26 | 24 |
| Participant 7 | 23 | 20 |
| Participant 8 | 20 | 22 |
| Participant 9 | 11 | 15 |
| Participant 10 | 17 | 16 |

The average score of the participants before the session was 22.5 and after one month was 22.3 with no significant difference.

Pittsburgh Sleep quality index:

| Participant | Pre | Post |
|---|---|---|
| Participant 1 | 13 | 9 |
| Participant 2 | 9 | 10 |
| Participant 3 | 14 | 10 |
| Participant 4 | 14 | 13 |
| Participant 5 | 7 | 1 |
| Participant 6 | 15 | 8 |
| Participant 7 | 2 | 0 |
| Participant 8 | 5 | 8 |
| Participant 9 | 4 | 0 |
| Participant 10 | 8 | 5 |

The average scores of the participants before the session was 9.1 and after one month was 6.4. While the difference appears significant - there was no control for variables during the one month due to the varying quality of sleep & participant pool.

**Study Planning Take-aways on how to conduct research in the Sonorium**

The study identified several areas for improvement and suggested changes to enhance the overall participant experience and study efficiency. We are at the moment devising a study protocol that facilitates a smooth planning of the study taking into account all what we have learned regarding the use of measurement tools, recruitment & selection of participants, overall logistical organization of the study and the analysis and organization of the collected data.

Additionally due to the varying nature of perceived stress and sleep quality - as it can be influenced by many other variables which cannot be controlled for - more robust post study assessment is recommended with interviews every 4 days and questionnaires every week for the next 2 weeks after the session. This calls for a more detailed follow up with the subjects to ensure an individualized approach for each case.

## Conclusion:

The discussion of the results is pending; however, it can already be stated that due to the limited number of subjects, no clear conclusions regarding the effect of the Sonorium can be drawn. As mentioned at the outset, this study is designed to define the future directions of subsequent studies as well as to establish and test the practical foundations for a study in the Sonorium environment. This approach underscores the exploratory nature of the research, aimed at laying the groundwork for more comprehensive investigations in the field.

## Further possible research:

- geared to a specific segment of the population: e.g. people aged 60-80
- the interaction of the specific instruments in the session. The facilitators reporting the sound bed and sonic stone reacting differently with different participants.

- comparison of population with no experience of meditation and those with experience of meditation
- sleeping problem research
- more EEG analysis with more segmentation of the population